Protocolos de comunicaciones de alta velocidad

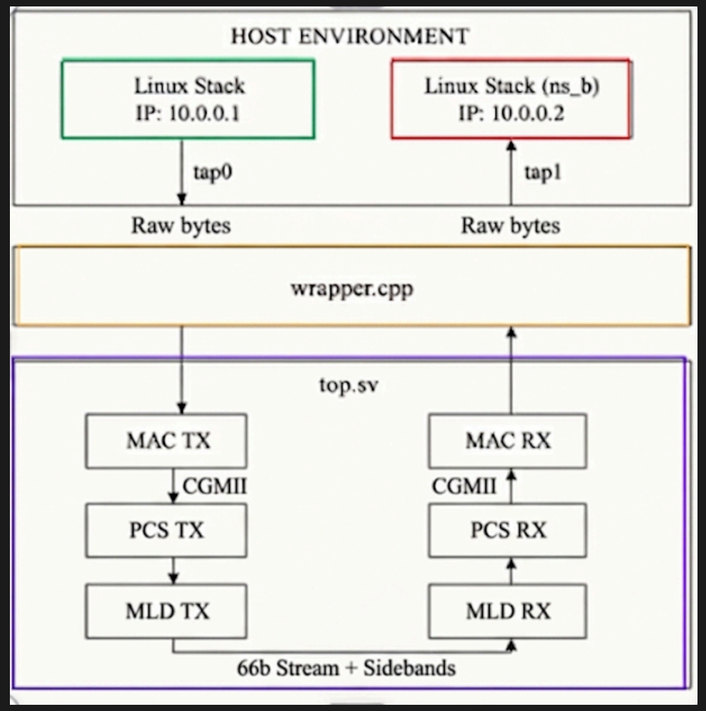
Ethernet-phy

Lab4 PCS Multi-Lane Distribution (MLD)

Fundación Fulgor

Facundo Guerrero

Architecture Setup



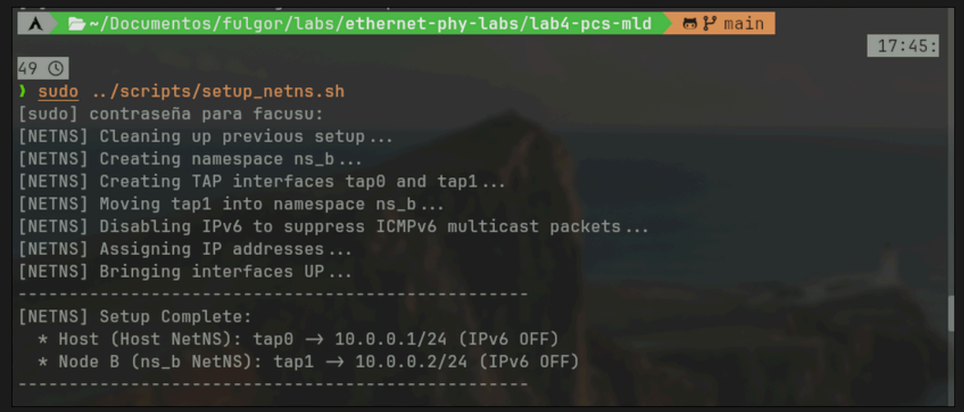
Multi-Lane Distribution (MLD) Overview

It is a mechanism that takes a high-speed stream and distributes the 66-bit blocks round-robin across multiple parallel lanes to make physical transmission feasible.

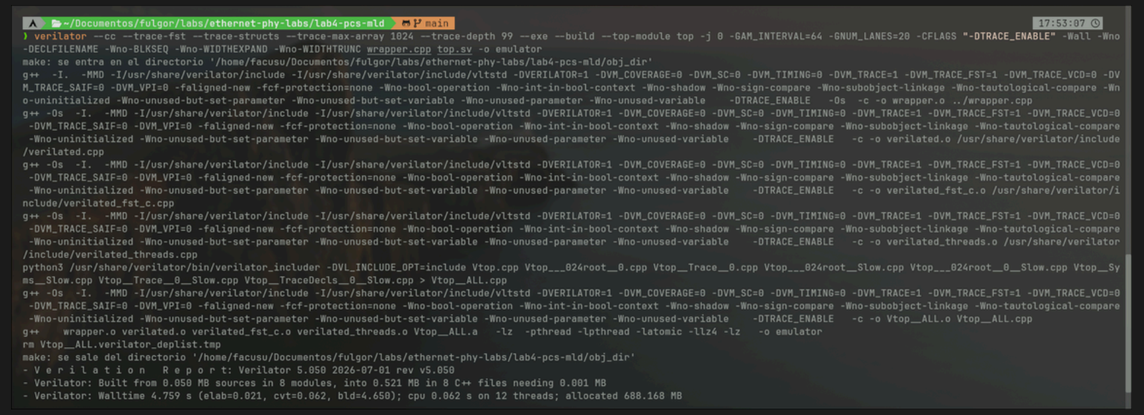
The stream is physically split into 20 distinct hardware lanes. To correct physical cable delays (skew) and rebuild the data, Alignment Markers (AMs) are periodically injected directly into each lane's data path.

Lab Adaptation: To simplify simulation, the data travels in a single unified stream. The 20 physical paths are abstracted into Virtual Lanes using a sideband tag (mld_lane_id), and AMs are indicated by a parallel pulse (mld_is_am).

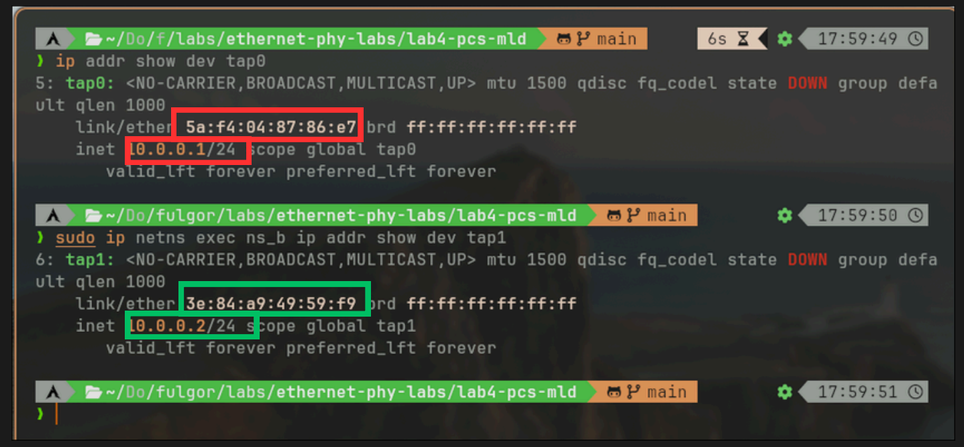
Architecture Setup



1. We run *setup_netns.sh* to create the TAPS



2. We use verilator to compile and generate an executable taking wrapper and top.sv files. This create an emulation executable and a dump file for GTK wave.

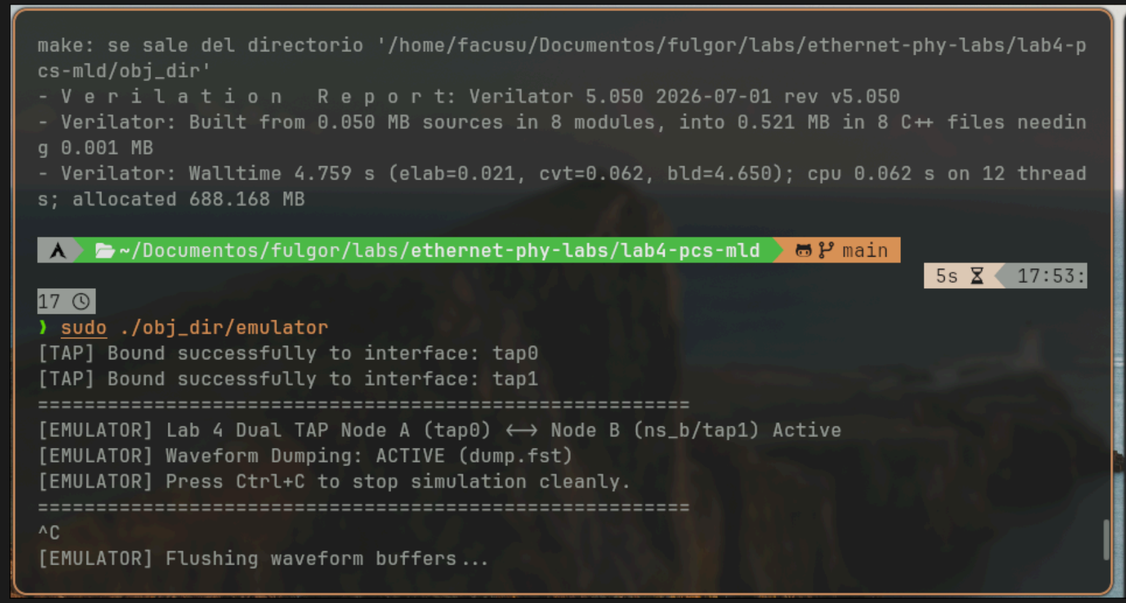


3. Finally there are the two taps and its two MACs addresses and ips

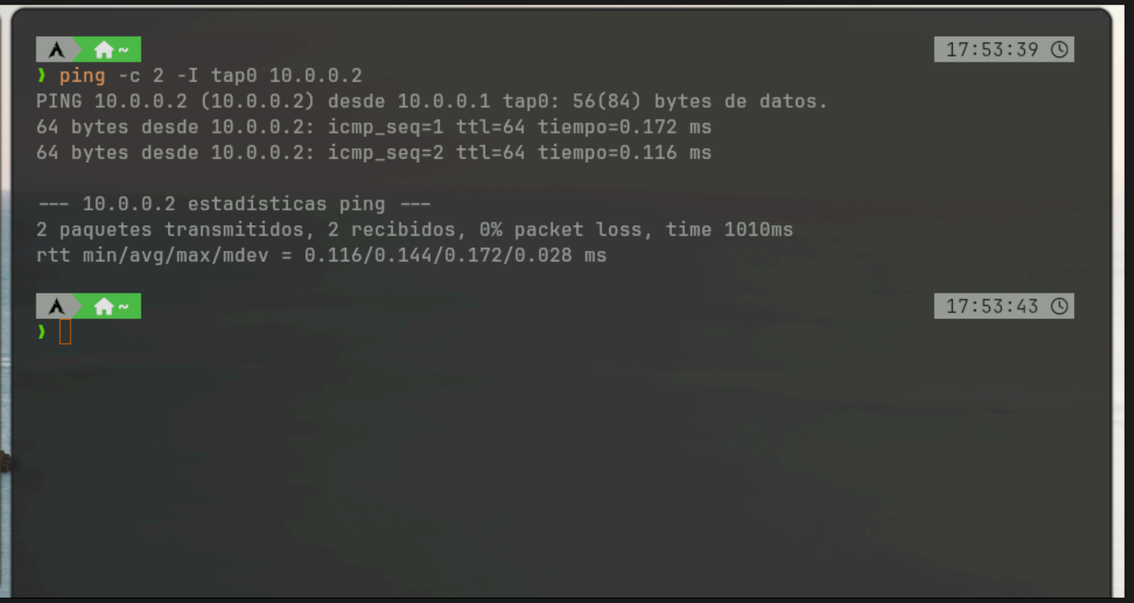
TAP	IP	MAC
Tap0	**10.0.0.1**	**5a:f4:04:87:86:e7**
Tap1	**10.0.0.2**	**3e:84:a9:49:59:f9**

Execution

It is executed simultaneously

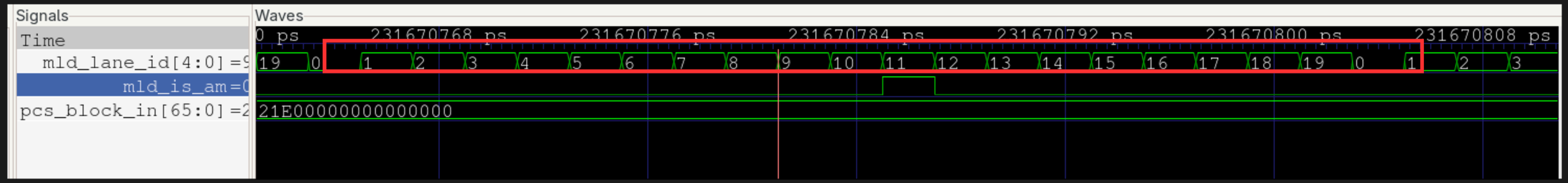


Simulation excecution



Ping command send

Analysis

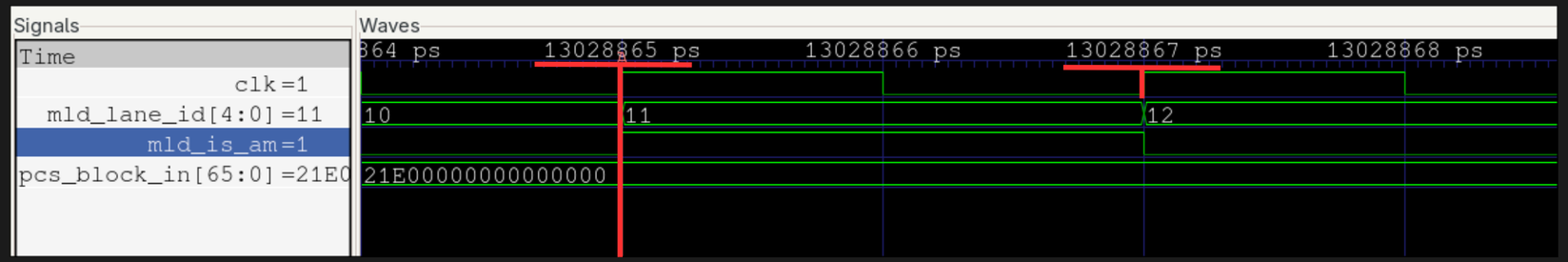


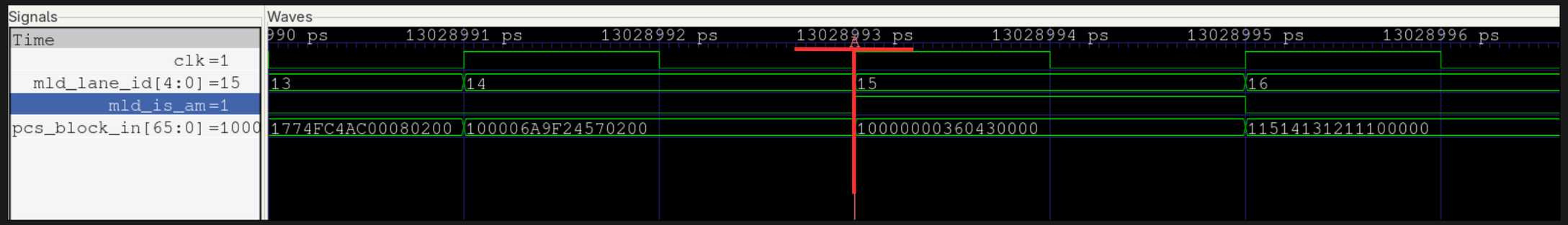
- We can see *how mld_lane_id* increment sequentially from 0 to 19 and then back to 0.



The imagen shows how *mld_lane_id* increments and a new block is dispatched, we can see datablock (destiny mac address) after preamble

Analysis

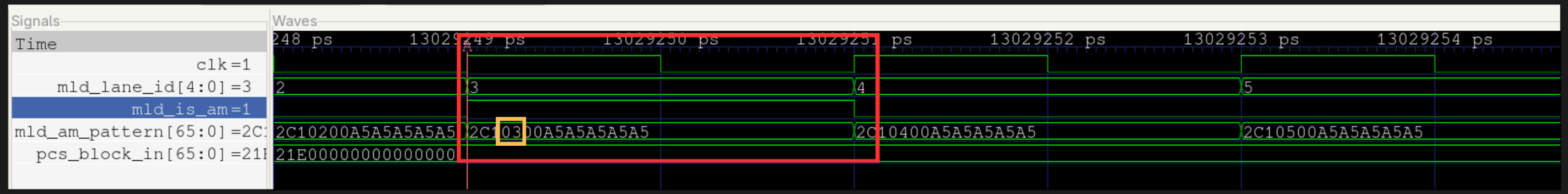


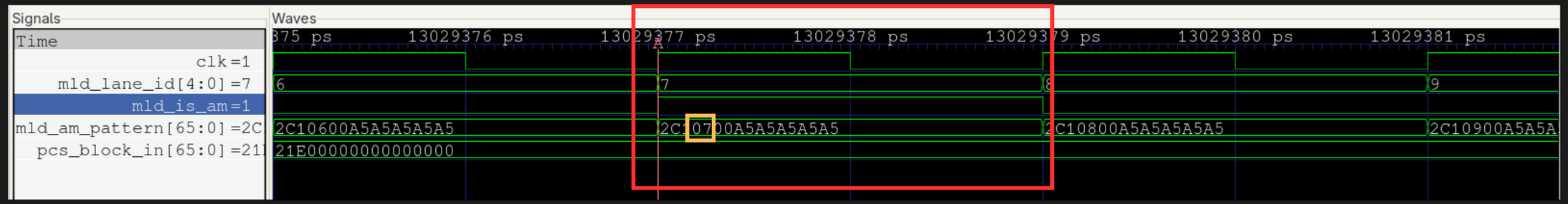


Time between two AM pulses

- We can see that one clock cycle occurs every 2 ps *(13028867ps - 13028865ps = 2ps).*
- If we compare the two images, we can see that *mld_is_am* is set to 1 (indicating an AM pulse) first at 13028865ps and then at the time instant 13028993 ps. (*13028993 ps - 13028865 ps = 128 ps).*
- Since each clock cycle occurs every 2 ps (*128 / 2 = 64 clock cycles.)*
- *S*ince each block is dispatched every clock cycle (we can see that *mld_lane_id* increases every clock cycle), this means that the period between AM pulses is **64 blocks**, which matches the simulation parameter **-GAM_INTERVAL=64.**

Analysis





- **We can see that the *mld_am_pattern* signal indicates the AM block being sent when the mld_is_am signal is set to 1.**
- **We can also notice the field that changes depending on the lane to which the AM pulse corresponds.**